

[Pranesh S A 251801234-2025-AIDS](#) ▾**P2****Started on** Thursday, 24 September 2026, 8:05 AM**State** Finished**Completed on** Thursday, 24 September 2026, 8:17 AM**Time taken** 11 mins 55 secs**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 10.00 out of 10.00**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3. Write any efficient algorithm to find the possible ways.

Example 1:**Input:** 6**Output:** 6**Explanation:** There are 6 ways to 6 represent number with 1 and 3

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format**Print:** The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d",&n);
6
7     if (n < 0)
8     {
9         printf("0\n");
10        return 0;
11    }
12    if (n == 0)
13    {
14        printf("1\n");
15        return 0;
16    }
17    if (n == 1 || n == 2)
18    {
19        printf("1\n");
20        return 0;
21    }
22    if (n == 3)
23    {
24        printf("2\n");
25        return 0;
26    }

```

```
26      f
27
28      long long w3 = 1;
29      long long w2 = 1;
30      long long w1 = 2;
31      long long current = 0;
32
33      for (int i = 4; i <= n; i++) {
34          current = w1 + w3;
35          w3 = w2;
36          w2 = w1;
37          w1 = current;
38      }
39
40      printf("%lld\n", w1);
41      return 0;
42 }
```

	Input	Expected	Got	
✓	6	6	6	✓
✓	25	8641	8641	✓
✓	100	24382819596721629	24382819596721629	✓

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

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